

Question ID: 4eaf0a3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $\sqrt[7]{x^9y^9}$, where x and y are positive?

Answer

- A. $(xy)^{\frac{7}{9}}$
- B. $(xy)^{\frac{9}{7}}$
- C. $(xy)^{16}$
- D. $(xy)^{63}$

Correct Answer: B

Rationale

Choice B is correct. For positive values of a and b , $a^mb^m = (ab)^m$, $\sqrt[n]{a} = (a)^{\frac{1}{n}}$, and $(a^j)^k = a^{jk}$. Therefore, the given expression, $\sqrt[7]{x^9y^9}$, can be rewritten as $\sqrt[7]{(xy)^9}$. This expression is equivalent to $\left((xy)^9\right)^{\frac{1}{7}}$, which can be rewritten as $(xy)^{9\cdot\frac{1}{7}}$, or $(xy)^{\frac{9}{7}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.